

GÖZÜMLER

C.1) A matrisinin idempotent olması için $A^2 = A$ olmalıdır.

$$A^2 = \begin{bmatrix} 2 & -2 & -4 \\ -1 & k & 4 \\ 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 & -2 & -4 \\ -1 & k & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 4+2-4 & -4-2k+8 & -8-8+12 \\ -2-k+4 & 2+k^2-8 & 4+4k-12 \\ 2+2-3 & -2-2k+6 & -4-8+9 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4-2k & -4 \\ 2-k & k^2-6 & 4k-8 \\ 1 & 4-2k & -3 \end{bmatrix} = \begin{bmatrix} 2 & -2 & -4 \\ -1 & k & 4 \\ 1 & -2 & -3 \end{bmatrix} = A$$

$$4-2k = -2$$

$$2k = 6$$

$$\textcircled{3} \quad \underline{k = 3}$$

C2)

$$\begin{vmatrix} 0 & x & y & z \\ x & 0 & z & y \\ y & z & 0 & x \\ z & y & x & 0 \end{vmatrix} = \begin{vmatrix} x+y+z & x & y & z \\ x+y+z & 0 & z & y \\ x+y+z & z & 0 & x \\ x+y+z & y & x & 0 \end{vmatrix}$$

$$= (x+y+z) \begin{vmatrix} 1 & x & y & z \\ 1 & 0 & z & y \\ 1 & z & 0 & x \\ 1 & y & x & 0 \end{vmatrix}$$

1. satırı (-1) ile çarpıp diğer satırlara ekleyelim.

$$= (x+y+z) \begin{vmatrix} 1 & x & y & z \\ 0 & -x & z-y & y-z \\ 0 & z-x & -y & x-z \\ 0 & y-x & x-y & -z \end{vmatrix}$$

1. sütuna göre açalım.

$$= (x+y+z) \begin{vmatrix} -x & z-y & y-z \\ z-x & -y & x-z \\ y-x & x-y & -z \end{vmatrix}$$

2. sütunu birinci ve üçüncü sütunlara ekleyelim.

$$= (x+y+z) \begin{vmatrix} z-(x+y) & z-y & 0 \\ z-(x+y) & -y & x-(y+z) \\ 0 & x-y & x-(y+z) \end{vmatrix}$$

$$= (x+y+z)(z-x-y)(x-y-z) \begin{vmatrix} 1 & z-y & 0 \\ 1 & -y & 1 \\ 0 & x-y & 1 \end{vmatrix}$$

$$= (x+y+z)(z-x-y)(x-y-z) \begin{vmatrix} 1 & z-y & 0 \\ 0 & -z & 1 \\ 0 & x-y & 1 \end{vmatrix}$$

$$= (x+y+z)(z-x-y)(x-y-z)[-z-(x-y)]$$

$$= (x+y+z)(z-x-y)(x-y-z)(y-x-z) \quad (5)$$

$$= -(x+y+z)(z-x-y)(-z^2 - (x-y)(y-x-z))$$

$$(5) \quad \langle p(t), q(t) \rangle = \int_0^1 p(t) q(t) dt$$

$$(1) \quad \langle p(t), p(t) \rangle \geq 0 \quad \langle p(t), p(t) \rangle = 0 \Leftrightarrow p(t) = 0$$

$$\int_0^1 |p(t)|^2 dt \geq 0. \quad \langle p(t), p(t) \rangle = 0 \Leftrightarrow p(t) = 0.$$

$$\int_0^1 \underline{|p(t)|^2} dt = c \Big|_0^1$$

(10)

$$(2) \quad \langle p(t), q(t) \rangle = \langle q(t), p(t) \rangle$$

$$\int_0^1 p(t) q(t) dt = \int_0^1 q(t) p(t) dt$$

(5)

$$(3) \quad k \langle p(t), q(t) \rangle = \langle k p(t), q(t) \rangle \quad (k \text{ sct})$$

$$k \int_0^1 p(t) q(t) dt = \int_0^1 \underline{k p(t)} q(t) dt = \langle k p(t), q(t) \rangle$$

(5)

$$(4) \quad \langle p(t) + q(t), w(t) \rangle = \langle p(t), w(t) \rangle + \langle q(t), w(t) \rangle$$

$$\int_0^1 (p(t) + q(t)) w(t) dt = \int_0^1 p(t) w(t) dt + \int_0^1 q(t) w(t) dt$$

$$= \int_0^1 \{ \underline{p(t) w(t) + q(t) w(t)} \} dt$$

(5)

$$2) \begin{bmatrix} 1 & 2 & -3 & 4 \\ 3 & -1 & 5 & 2 \\ 4 & 1 & k-14 & k+2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & -7 & k-2 & k-14 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & 0 & k-16 & k-4 \end{bmatrix} \quad \text{a) } \begin{matrix} k^2-16=0 & k-4 \neq 0 \\ k=-4 & k \neq 4 \end{matrix} \quad (5)$$

$k=-4$ olmalıdır. ($k=-4$ ise çözüm yoktur)

b) $k \neq -4$ ise tek çözüm vardır. (5)

c) $k=4$ ise sonsuz çözüm vardır. (5)

$$\boxed{c_1 \vec{a} + c_2 \vec{b} + c_3 \vec{c} = 0} \quad (1)$$

3)
$$\begin{vmatrix} 2m & -1 & -1 \\ -1 & 2m & -1 \\ -1 & -1 & 2m \end{vmatrix} = 0$$
 abmaldiver 5

$$\begin{vmatrix} 2m-2 & 2m-2 & 2m-2 \\ -1 & 2m & -1 \\ -1 & -1 & 2m \end{vmatrix} = \begin{vmatrix} 2m-2 & 0 & 0 \\ -1 & 2m+1 & 0 \\ -1 & 0 & 2m+1 \end{vmatrix}$$

$$= (2m-2)(2m+1)^2 = 0$$

$m_1 = 1, m_{2,3} = -\frac{1}{2}$ *abmaldiver* 10

$m_1 = 1$ ist $c_1 \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -2 & 0 \\ 0 & 3 & -3 & 0 \\ 0 & -3 & 3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\vec{a} + \vec{b} + \vec{c} = 0}$$
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$$\begin{cases} c_1 - c_2 = 0 \\ c_2 - c_3 = 0 \end{cases} \Rightarrow \begin{cases} c_1 = s \\ c_2 = s \\ c_3 = s \end{cases} \quad (s=1 \text{ ist m.})$$

oder $c_2 = 1$
 $c_1 = 1$
 $c_3 = 1$

$m_{2,3} = -\frac{1}{2}$ ist:

$$\begin{bmatrix} -1 & -1 & -1 & 0 \\ -1 & -1 & -1 & 0 \\ -1 & -1 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{-2\vec{a} + \vec{b} + \vec{c} = 0}$$

5

$$c_1 + c_2 + c_3 = 0$$

$c_2 = 1$ kept
 $c_3 = 1$ kept
 $c_1 = -2$ *oder*

$$4) A = \begin{bmatrix} 1 & -4 & 2 \\ -4 & 1 & -2 \\ 2 & -2 & -2 \end{bmatrix} \quad (A - \lambda I)X = 0$$

$$\begin{vmatrix} 1-\lambda & -4 & 2 \\ -4 & 1-\lambda & -2 \\ 2 & -2 & -2-\lambda \end{vmatrix} = 0$$

$$|A - \lambda I| = 0 \quad (5)$$

$$\begin{cases} (1-\lambda)x_1 - 4x_2 + 2x_3 = 0 \\ -4x_1 + (1-\lambda)x_2 - 2x_3 = 0 \\ 2x_1 - 2x_2 + (-2-\lambda)x_3 = 0 \end{cases}$$

$$(1-\lambda)[(1-\lambda)(-2-\lambda)-4] + 4[8+4\lambda+4] + 2[8-2+2\lambda] = 0$$

$$\begin{vmatrix} 1-\lambda & -4 & 2 \\ -4 & 1-\lambda & -2 \\ 2 & -2 & -2-\lambda \end{vmatrix} = \begin{vmatrix} -3-\lambda & -4 & 2 \\ -3-\lambda & 1-\lambda & -2 \\ 0 & -2 & -2-\lambda \end{vmatrix} = \begin{vmatrix} -3-\lambda & -4 & 2 \\ 0 & 5-\lambda & -4 \\ 0 & -2 & -2-\lambda \end{vmatrix}$$

$$= (-3-\lambda)[\lambda^2 - 3\lambda - 18] = 0$$

$$(1-\lambda)[\lambda^2 + \lambda - 6] + 16\lambda + 48 + 4\lambda + 12 = 0$$

$$\lambda^2 + \lambda - 6 - \lambda^3 - \lambda^2 + 6\lambda + 20\lambda + 60 = 0$$

$$-\lambda^3 + 27\lambda + 54 = 0$$

$$\lambda_1 = 6, \lambda_2 = -3, \lambda_3 = -3$$

$$\lambda_1 = 6, \lambda_2 = -3, \lambda_3 = -3 \text{ bulunur. } (10)$$

$$\lambda_1 = 6 \text{ için } \begin{cases} -5x_1 - 4x_2 + 2x_3 = 0 \\ -4x_1 - 5x_2 - 2x_3 = 0 \\ 2x_1 - 2x_2 - 8x_3 = 0 \end{cases} \quad \begin{bmatrix} -5 & -4 & 2 & 0 \\ -4 & -5 & -2 & 0 \\ 2 & -2 & -8 & 0 \end{bmatrix} \sim$$

$$\sim \begin{bmatrix} 1 & -1 & -4 & 0 \\ 0 & -9 & -18 & 0 \\ 0 & -9 & -18 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{cases} x_1 - 2x_3 = 0 \\ x_2 + 2x_3 = 0 \\ x_3 = 25 \end{cases}$$

$$X_1 = 5 \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \quad (5)$$

$$\lambda_{2,3} = -3 \text{ için } \begin{cases} 4x_1 - 4x_2 + 2x_3 = 0 \\ -4x_1 + 4x_2 - 2x_3 = 0 \\ 2x_1 - 2x_2 + x_3 = 0 \end{cases}$$

$$\begin{bmatrix} 4 & -4 & 2 & 0 \\ -4 & 4 & -2 & 0 \\ 2 & -2 & 1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$X_{2,3} = s \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{bulunur } (5)$$

$$x_1 - x_2 + \frac{1}{2}x_3 = 0$$

$$\begin{cases} x_3 = 25 \\ x_2 = t \\ x_1 = t - 5 \end{cases} \text{ bulunur}$$

5) a) 1) $\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = 0$

2) $\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = 0 \Leftrightarrow \alpha_1 = \alpha_2 = \alpha_3 = 0$

Her iki grupta da çözümü olmalı, 2. homojen sistemin tek çözümü olmalı

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$$\begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & 2 \\ 1 & 0 & 1 \end{vmatrix} = -1 \neq 0 \text{ tek çözüm. Dolayısıyla bizzat.}$$

b) $u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad u_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \quad u_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \rightarrow \text{ortogonal baz.}$

$v_1 = u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad v_2 = u_2 - \frac{(u_2, v_1)}{(v_1, v_1)} v_1 = u_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

$v_3 = u_3 - \frac{(u_3, v_2)}{(v_2, v_2)} v_2 - \frac{(u_3, v_1)}{(v_1, v_1)} v_1$

$= \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - \frac{4}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 1 + 1/2 - 4/3 \\ 2 - 1/2 - 4/3 \\ 1 - 0 - 4/3 \end{bmatrix} = \begin{bmatrix} 3/2 - 4/3 \\ 3/2 - 4/3 \\ -2/6 \end{bmatrix} = \begin{bmatrix} 1/6 \\ 1/6 \\ -2/6 \end{bmatrix}$

$w_1 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix} \quad w_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \quad w_3 = \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ -2/\sqrt{6} \end{bmatrix}$

$$(1) \quad A = \begin{bmatrix} 1 & -2 & 1 \\ 1 & 3 & 5 \end{bmatrix} \quad X = \begin{bmatrix} x & a \\ y & b \\ z & c \end{bmatrix}$$

$$AX = \begin{bmatrix} x-2y+z & a-2b+c \\ x+3y+5z & a+3b+5c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad (5)$$

$$\begin{aligned} x-2y+z &= 1 \\ x+3y+5z &= 0 \end{aligned} \quad \left[\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 1 & 3 & 5 & 0 \end{array} \right] \sim \left[\begin{array}{ccc|c} 0 & -5 & -4 & 1 \\ 1 & 3 & 5 & 0 \end{array} \right]$$

$$x_3 = t \quad x_2 = \frac{1+4t}{-5}$$

$$x_1 + 3x_2 + 5x_3 = 0$$

$$x_1 + \frac{3+12t}{-5} + 5t = 0$$

$$x_1 = -\frac{13t+3}{5}$$

(10)

$$a-2b+c=0$$

$$0=2b-c$$

$$a+3b+5c=1$$

$$5b+4c=1$$

$$c = \frac{1-5b}{4}$$

$$b = \frac{1-4c}{5}$$

$$a = \frac{2-8t}{5} - t$$

$$a = \frac{2-13t}{5}$$

(10)

$$X = \begin{bmatrix} \frac{-13t+3}{5} & \frac{2-13t}{5} \\ \frac{1+4t}{-5} & \frac{1-4t}{5} \\ t & t \end{bmatrix}$$

$$X = \begin{bmatrix} +3/5 & 2/5 \\ -1/5 & 1/5 \\ 0 & 0 \end{bmatrix}$$

$$t=1 \quad s=1$$